

Viyom Shukla



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🔗 TUF+ (Solved 300+ DSA problems) 🌐 viyomshukla

👉 LeetCode(200+ problems) ➤ GeeksforGeeks(100+ problems) 🌐 Portfolio

Education

Jaypee Institute of Information Technology, Noida

2022 – 2027

Integrated B.Tech + M.Tech in Computer Science Engineering – CGPA: 7.62/10

Technical Skills

Programming Languages: C, C++, JavaScript, Python, HTML, CSS, SQL

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS

Tools & Technologies: Git, GitHub, MongoDB, MySQL, PostgreSQL, VS Code, REST APIs, Notion, Slack

Core Competencies: Data Structures & Algorithms, Object-Oriented Design, Problem Solving, Complexity Analysis, Distributed Systems, Scalable Systems, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Data Mining

Experience

Web Developer Intern

Aug 2025 – Oct 2025

Distance Connect

- Designed and developed scalable web application using React.js and Node.js in a multi-tier architecture.
- Built and integrated APIs enabling efficient data processing and real-time communication in distributed systems.
- Reduced frontend load time by 40% through lazy loading, memoization, and component-level code splitting.

Publications

Enhancing Network Intrusion Detection Using Conditional GANs for Imbalanced Data Classification

IEEE Xplore Digital Library 🔗

- Divided NSL-KDD dataset into 4 binary classification subsets (Normal vs DoS, Probe, R2L, U2R) for targeted attack detection in embedded network security systems.
- Generated synthetic data using Conditional GANs for minority attack classes (Probe, R2L, U2R) achieving 99.8% accuracy on imbalanced datasets.
- Integrated ensemble learning methods (Random Forest, SVM, Neural Networks) demonstrating optimal performance across different attack types for real-time threat detection.

Projects

Thumblify: AI-Powered SaaS (Distributed System Integration)

Source Code 🔗

- Built scalable MERN SaaS with REST APIs, JWT auth, and fault-tolerant image processing for 100+ users.
- Engineered a scalable MERN platform using APIs (Gemini, Replicate) for real-time, fault-tolerant image processing.
- Built a low-cost WhatsApp bot via Twilio API, enabling users to generate AI thumbnails via mobile.
- Developed a Chrome extension and MERN backend optimized for high-speed performance and incredible scale.

DeepMood: Depression Prediction Using GANs

Notebook 🔗

- Developed machine learning model using Generative Adversarial Networks (GANs) achieving 92% accuracy in depression prediction.
- Trained model on mental health datasets and generated synthetic data to address data scarcity in clinical datasets.

NextTrip: Tour Booking Management Platform

Source Code 🔗

- Engineered full-stack MERN application with role-based authentication for secure admin and user access.
- Designed admin dashboard for real-time tour management and booking analytics with MongoDB database integration.
- Designed database schema and optimized queries ensuring efficient storage and retrieval.

WalletWatcher: Expense Management Web App

Live Link 🔗

- Engineered a responsive MERN dashboard to visualize spending habits and manage personal budgets..
- Secured sensitive financial data using JWT-based authentication and industry-standard encryption.

Achievements

- International Conference Speaker:** Presented research on Network Intrusion Detection at 17th IC3-2025, IIIT Noida. *Certificate* 🔗
- Student Placement Committee Member:** Coordinating placement activities and supporting 200+ students in job search.
- Hackathon Experience:** Participated in premier national hackathons including Smart India Hackathon 2023, Adobe India Hackathon, Flipkart Grid 6.0, and Amazon ML Challenge, solving real-world problems.
- LeetCode Knight:** 350+ problems solved across Arrays, Trees, Graphs, and DP on LeetCode and TakeUforward.